ONLINE EXAM PORTAL ? TESTING EVIDENCE

Combined Code and Branch Coverage Report

Backend and frontend Istanbul-compatible coverage evidence

Repository: SE-Team-6/online_exam_portal

Branch: main **Commit:** 2914e5c55a79108e5b24fbb9bdd181fffbf119a1

Evidence date: 10 November 2026

Scope: Results and evidence collected for the current main checkout; no full-system sign-off implied.

Prepared from actual automated test output, browser validation output, screenshots, and the main-branch CI coverage artifact. Untested items are explicitly identified.

Combined aggregate metrics

Metric	Backend covered / total	Frontend covered / total	Combined covered / total	Combined percentage
Statements	415 / 607 (undefined%)	124 / 597 (20.77%)	**539 / 1204**	**44.77%**

Metric	Backend covered / total	Frontend covered / total	Combined covered / total	Combined percentage
Branches	260 / 458 (undefined%)	75 / 513 (14.61%)	**335 / 971**	**34.5%**
Functions	81 / 112 (undefined%)	30 / 153 (19.6%)	**111 / 265**	**41.89%**
Lines	399 / 566 (undefined%)	110 / 545 (20.18%)	**509 / 1111**	**45.81%**

Combined percentages are calculated from the summed covered and total counts across both projects (not by averaging the two percentages). Backend statement coverage is reported here from the validated Istanbul conversion; Node?s native summary itself does not report statement coverage.

Evidence sources and method

- **Backend:** Node v24.19.0 built-in test coverage run with raw V8 data retained in a temporary directory. 109 valid raw coverage records were grouped by source URL, pairwise merged, and converted using the installed AST-aware `ast-v8-to-istanbul` package into Istanbul file-coverage counters. 35 backend source files were converted; per-file counters were summed and verified against the aggregate totals below.
- **Frontend:** CI run #111, successful on main commit `2914e5c55a79108e5b24fbb9bdd181fffbf119a1`. Artifact `frontend-coverage` (ID 13742368873; 154,891 bytes), produced by Vitest 4.1.12?s V8 coverage provider. CI recorded 4 test files and 12 passing tests. Artifact expiry was listed as 2027-02-08.
- **Metric compatibility:** Both sides are expressed as Istanbul statement, branch, function, and line counters: frontend from Vitest?s V8/Istanbul coverage artifact, backend from raw V8 records converted with AST-aware V8-to-Istanbul. The denominators are summed directly. This supports an aggregate count-based comparison, while source scopes and execution suites remain distinct.

Backend per-file validated totals (35 files)

Backend source file	Statements	Branches	Functions	Lines
`backend/src/app.js`	21/22	2/4	0/0	21/21
`backend/src/routes/health.routes.js`	3/3	0/0	1/1	3/3
`backend/src/middleware/requestId.js`	3/3	0/0	1/1	3/3
`backend/src/middleware/notFound.js`	4/4	0/0	1/1	4/4
`backend/src/middleware/errorHandler.js`	3/4	6/11	1/1	3/4
`backend/src/middleware/cors.js`	5/11	4/6	1/1	5/11
`backend/src/routes/adminQuestion.routes.js`	12/21	0/0	2/7	12/21
`backend/src/middleware/asyncHandler.js`	3/3	0/0	2/2	2/2
`backend/src/services/question.service.js`	28/51	26/38	9/15	26/49
`backend/src/models/Questions.js`	7/8	1/2	3/3	7/8
`backend/src/middleware/requireAuth.js`	9/10	6/6	2/2	8/9
Validated backend total	**415/607**	**260/458**	**81/112**	**399/566**

Backend source file	Statements	Branches	Functions	Lines
backend/src/utils/jwt.js	10/13	18/22	3/3	10/12
backend/src/middleware/requireRole.js	5/6	3/4	2/2	5/6
backend/src/routes/auth.routes.js	33/46	17/32	3/4	33/46
backend/src/models/User.js	1/1	0/0	0/0	1/1
backend/src/utils/password.js	14/15	13/14	2/2	14/15
backend/src/middleware/requireHttps.js	4/6	6/11	1/1	4/6
backend/src/routes/audit.routes.js	3/14	0/14	0/1	3/11
backend/src/models/AuditLog.js	1/1	0/0	0/0	1/1
backend/src/middleware/auditEvents.js	29/31	42/54	4/5	27/29
backend/src/services/audit.service.js	2/3	9/12	1/1	2/2
backend/src/routes/attempt.routes.js	15/17	0/0	4/5	15/17
backend/src/services/grading.service.js	21/50	14/34	5/8	18/36
backend/src/models/Attempt.js	3/3	0/0	0/0	3/3
backend/src/models/Exam.js	1/1	0/0	0/0	1/1
backend/src/models/Result.js	1/1	0/0	0/0	1/1
backend/src/services/attempt.service.js	108/150	71/116	17/17	107/149
backend/src/routes/examAttempt.routes.js	5/5	0/0	1/1	5/5
backend/src/routes/adminExam.routes.js	4/8	0/0	0/2	4/8
backend/src/services/exam.service.js	0/29	0/34	0/8	0/24
backend/src/routes/exam.routes.js	4/6	0/0	0/1	4/6
backend/src/services/studentExam.service.js	0/3	0/3	0/2	0/2
backend/src/routes/studentResults.routes.js	5/5	0/0	1/1	5/5
backend/src/services/reporting.service.js	43/48	22/41	13/13	37/40
backend/src/routes/adminResults.routes.js	5/5	0/0	1/1	5/5
Validated backend total	**415/607**	**260/458**	**81/112**	**399/566**

Native Node summary comparison and limitations

The same backend test run?s Node-native summary reported 76.12% lines, 69.65% branches, and 76.42% functions; it did not report statement counts. Those percentages differ from the custom Istanbul conversion (70.49%, 56.76%,

and 72.32%, respectively). Node?s built-in summary applies its own V8 range-to-line/branch/function reporting and aggregation, whereas this extraction maps raw V8 ranges through AST-aware conversion to Istanbul counters. Metric mapping and denominators differ, so the two summaries are not interchangeable. This report uses only the validated converted backend covered/total counts for the combined Istanbul-compatible aggregation and preserves Node?s native numbers as a separate reference.

Frontend artifact coverage: 124/597 statements, 75/513 branches, 30/153 functions, 110/545 lines. Backend conversion validation: 35 files, 0 invalid records, all four per-file sums matched the specified aggregates.